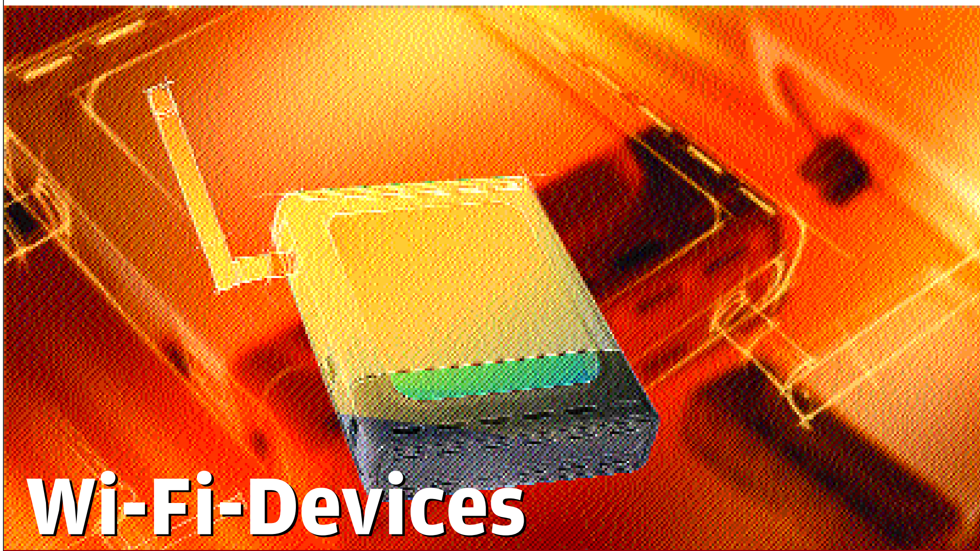

Wi-Fi-Devices

802.11g is already here pumping 54 Mbps of data which is higher than 11 Mbps offered by 802.11b. But soon 54 Mbps data transfer will fall short of expectation. Therefore IEEE has already started work on 108 Mbps and 300 Mbps.

Althought official names haven't been given to these standards but tentatively will be called 802.11n. 802.11i is also under development but this standard will address security issues with the WPA -there won't be any speed boost with this standard.

802.11h and 802.11j are another standards under scrutiny, which will address regional frequency-regulation requirements of Europe and Japan respectively.

Do Remember

- If the number of users is more just single Access point is not going to be enough. In such scenario more than one access point will have to be used in bridge mode.
- Site survey is important especially for offices and multistorey homes before hooking up the access point and placing the Wi-Fi enabled PC or laptop. Check for the spot where the signal strength is best and place the devices accordingly.
- Thick curtains and walls can be a major block to the signal. Although line of site is not necessary for communication between two Wi-Fi devices—but lesser the hindrance better the performance.
- While selecting Wi-Fi devices make sure you products support both the WEP and WPA security protocol for a safer and secure wireless network.
- If you are on a shoe string budget look out for the those access points that have in-built firewall, three-four LAN ports etc. They are generally classified as router access points.
- Since there are lot of issues in setting up Wi-Fi than in standard LAN, Wi-Fi device vendors offer facilities, where they survey you site for proper installation, bank on such surveys for effective and scientific installation.

Buying Tips

- If you are planning to connect just two desktop PCs at home or office, than just buy to PCI based Wi-Fi cards and run them on Ad-Hoc basis.
- If more than two systems are to a wired network (Internet, or existing LAN) needs to be accessed by the wirelessly connected PCs, be connected wirelessly than an Access Point is required.
- Depending on the type of usage and number of users, select between 802.11b and 802.11g standards. The latter has faster theoretical transfer rate of 54 Mbps compared to 11Mbps of 802.11b standard.
- Buy both Access point and card from same manufacturer as far as possible for full compatibility. Most manufacturers are offering feature of faster data transfer than 802.11b supports with their 802.11b compliant hardware. But higher data transfer will only work between their own Access point Point and card, if card from another vendor is used, it will work at lower transfer rates.
- Check the build quality of the Antenna. Since it can be rotated and in most models can be replaced, good build quality is imperative. Also prefer a PCI card that has an optional antenna connected with the cable. This allows the flexibility of placing the antenna at the convenient location for best signal reception.
- Software that let you set IP and other related settings need to be simple and intuitive. Check out the software and go fort the one that's simple to use and doesn't involve long learning curve.
- Access point should have the slots for wall mount and rubber pegs for proper placement on the desk. It should also be aesthetically pleasing and light in weight.
- Check out the security option supported by the Wi-Fi hardware. It should support both WEP (Wired Equivalent Privacy) and WPA (Wi-Fi Protected Access). WPA is the improvement over WEP, and therefore, is more secure.

Ready Reckoner

Low-end You Need	Home networking of two or three PCs for Internet or file sharing
Look for	3 Wi-Fi cards working in Ad-Hoc basis
Mid-end You Need	At least 5 PC with internet sharing, file sharing that contains large files like images and so on.
Look for	An Access-point running in infrastructure mode.
High-end You Need	10 to 15 PCs spread over big area.
Look for	More than one Access point in Bridge mode with rest of the card connecting to either of the access points